

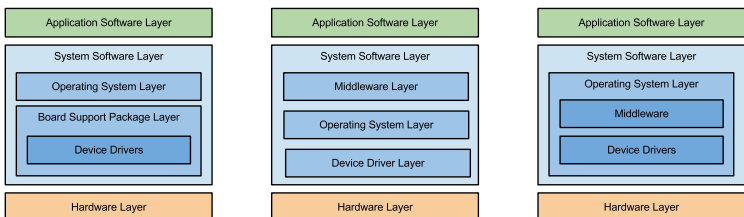
## Embedded software

Software running on embedded systems is vastly different from those that we run every day on our desktops and/or laptops. Since software in most cases is customised for the hardware on which it is to be deployed, software for embedded systems must be designed keeping in mind the limitation and capabilities of the platform. Embedded systems impose constraints of time and resources. Due to the timing constraints, embedded systems are often referred to as real-time systems i.e. the correctness of a computation not only depends on the logical correctness of the result but also the time at which the result is produced. The system is said to have failed if it doesn't meet the timing constraints. Resources on such systems can vary from 6KB ROM and 4KB RAM with a CPU operating at 8MHz to specifications that can rival modern computers but which are limited by the power available.

The eponymous operating systems that power these systems are designed to be compact, reliable and efficient at utilising the resources. Such OSES are not required in the case of small systems which can be easily controlled by a simple set of instructions stored in the device's ROM. However, they're a requirement for medium to large scale systems which must perform multiple tasks in real time. Often these operating systems are referred to as real-time operating systems (RTOS).

## Embedded / real-time operating systems

For an embedded system, the OS is a set of libraries which sits either over the hardware, over the device driver layer or the BSP (Board Support Package) layer. It acts as an abstraction layer between the hardware and the software running on top of the OS so that it doesn't have to deal with the hardware's idiosyncrasies. This also allows developers to come up with better software and middleware. It goes without saying that it has to do all these operations keeping the constraints associated with real-time systems in mind – it has to ensure timely completion of tasks. manage



Embedded OS Models